Supply Chain Analytics Dashboard

Project Report with Dashboard Images and Explanations

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1. Introduction

The Supply Chain Analytics Dashboard is an interactive Streamlit application for analysing supply chain performance across revenue, inventory, supplier quality, cost, demand, lead time, and profitability metrics. The dashboard combines business KPIs, interactive charts, statistical techniques, and machine learning models in one interface.

Users can move between multiple analytical sections using the sidebar and can apply filters for product type, location, and supplier. This makes the dashboard useful for both high-level performance review and detailed metric-level analysis.

2. Objectives

- Present supply chain KPIs in a clear and interactive dashboard.
- Analyse revenue trends, product contribution, supplier quality, and location-level inventory behaviour.
- Use box plots and IQR logic to identify variation and outliers.
- Apply regression analysis for prediction and relationship study.
- Compare simple random, systematic, and stratified sampling methods.
- Use K-Means clustering to identify meaningful operational segments.
- Study probability distributions and calculate interval probabilities for selected metrics.

3. Tools and Technologies

Technology	Purpose
Python	Core programming language
Streamlit	Web dashboard framework
Pandas	Data manipulation and tabular analysis
NumPy	Numerical computation
Plotly	Interactive charts
Scikit-learn	Linear regression and K-Means clustering
SciPy	Probability modelling and statistical testing

4. Dataset Fields

The dashboard works with supply chain records containing product, supplier, location, price, demand, revenue, stock, lead time, shipping cost, manufacturing cost, defect rate, profit margin, and month fields. These attributes allow analysis of commercial performance, operational efficiency, and quality performance.

5. Overview Dashboard

Figure 5: Overview Dashboard

Explanation

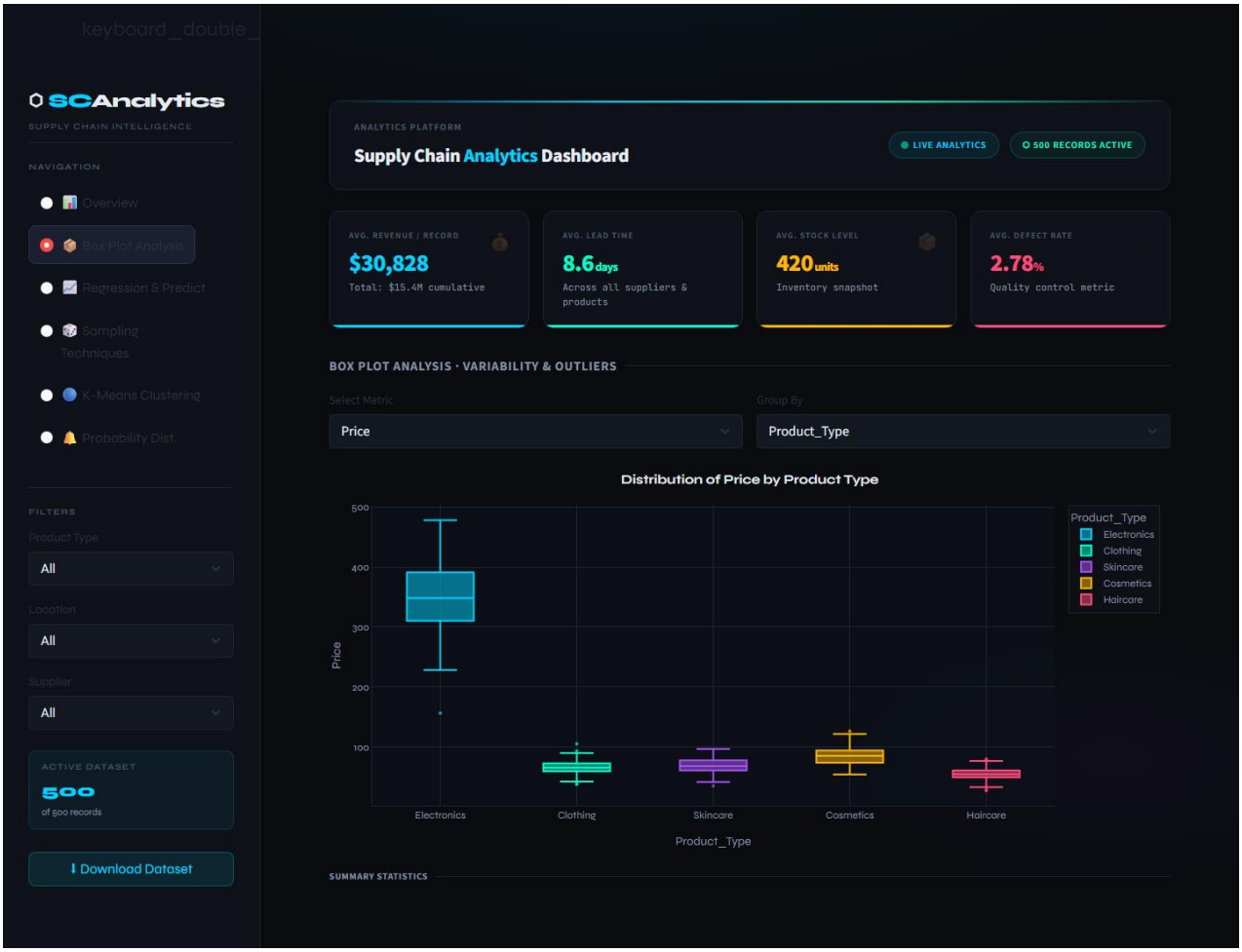
The Overview page is the main summary screen of the application. It combines KPI cards with visual charts to provide a quick understanding of supply chain performance.

The KPI cards show average revenue per record, average lead time, average stock level, and average defect rate. Below the KPIs, the dashboard visualizes monthly revenue, product revenue share, supplier defect rate, and location-level stock behaviour.

Key Observations

- The revenue trend chart helps compare product category performance over time.
- The donut chart highlights the relative revenue contribution of each product type.
- The supplier defect chart is useful for quality monitoring and supplier review.
- The location scatter plot connects revenue, demand, and stock level in one visual view.

6. Box Plot Analysis



SUMMARY STATISTICS

Product_Type	count	mean	std	min	25%	50%	75%	max
Clothing	100.000000	66.284000	11.934000	37.860000	59.598000	65.815000	72.755000	105.410000
Cosmetics	101.000000	85.557000	15.022000	54.350000	73.890000	85.500000	94.290000	126.470000
Electronics	106.000000	344.631000	60.146000	156.400000	310.642000	348.065000	390.700000	477.980000
Haircare	95.000000	54.828000	10.359000	27.500000	49.665000	54.720000	60.885000	79.960000
Skincare	98.000000	68.455000	11.849000	35.190000	61.055000	68.420000	77.958000	96.730000

MEAN	MEDIAN	STD DEV	OUTLIERS
127.435	73.610	117.023	106

Outlier Records — 106 detected

Box Plot Insight: Electronics shows the highest mean Price among all groups. The IQR method flagged **106 outlier records** — these warrant investigation as they may indicate data errors, exceptional events, or process anomalies in the supply chain.

Figure 6: Box Plot Analysis

Explanation

The Box Plot Analysis section is designed to compare the distribution of a selected numerical metric across a selected grouping variable. Users can choose metrics such as revenue, demand, price, stock level, lead time, shipping cost, manufacturing cost, defect rate, and profit margin.

The dashboard also calculates summary statistics and identifies outliers using the interquartile range method. This makes the module useful for detecting extreme values and understanding variability across product types, suppliers, or locations.

Key Observations

- Box plots show the median, spread, and possible outliers for each group.
- The summary table supports numeric comparison between groups.
- Outlier counts help identify unusual records that may need investigation.
- This view is especially helpful for quality, demand, cost, and revenue analysis.

7. Regression and Prediction

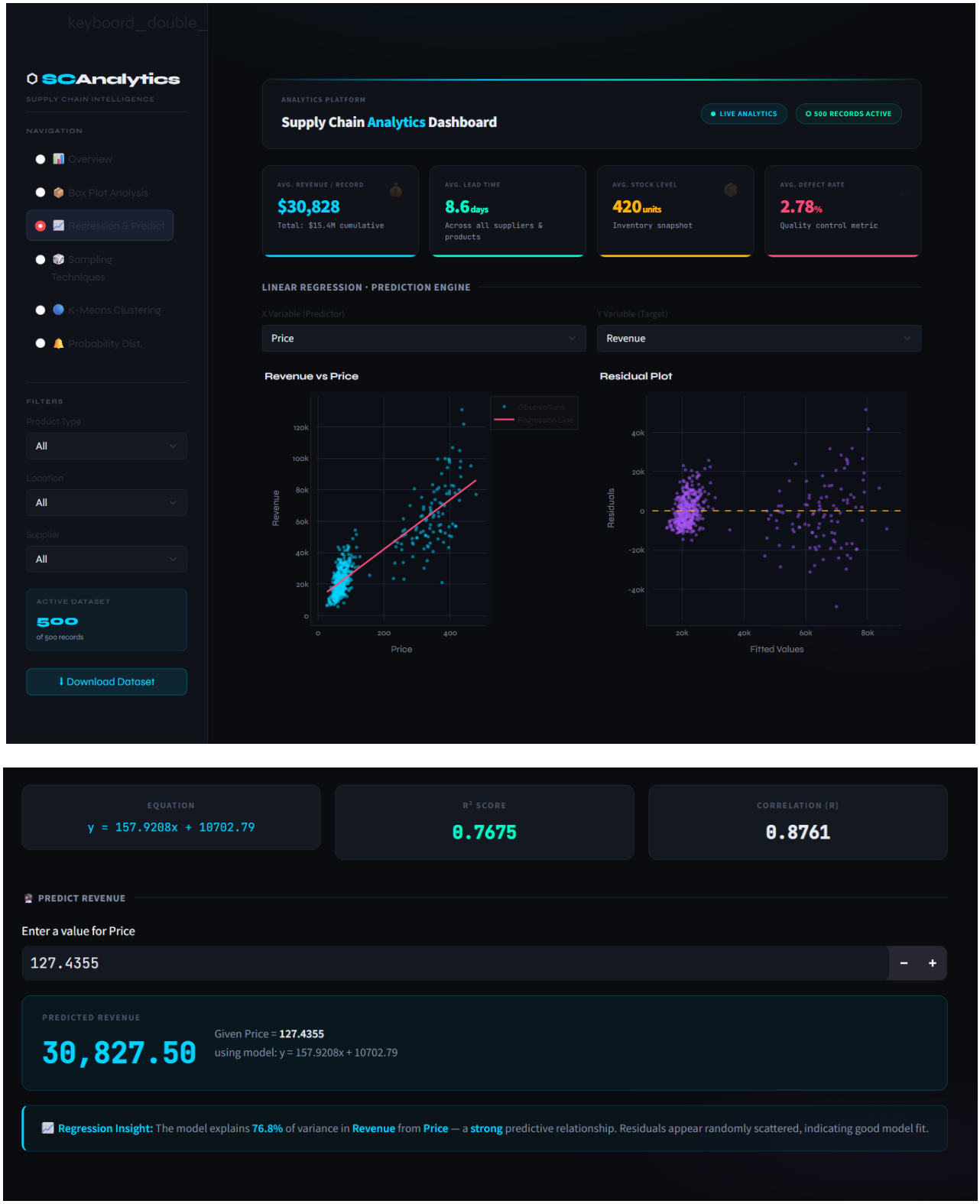


Figure 7: Regression and Prediction

Explanation

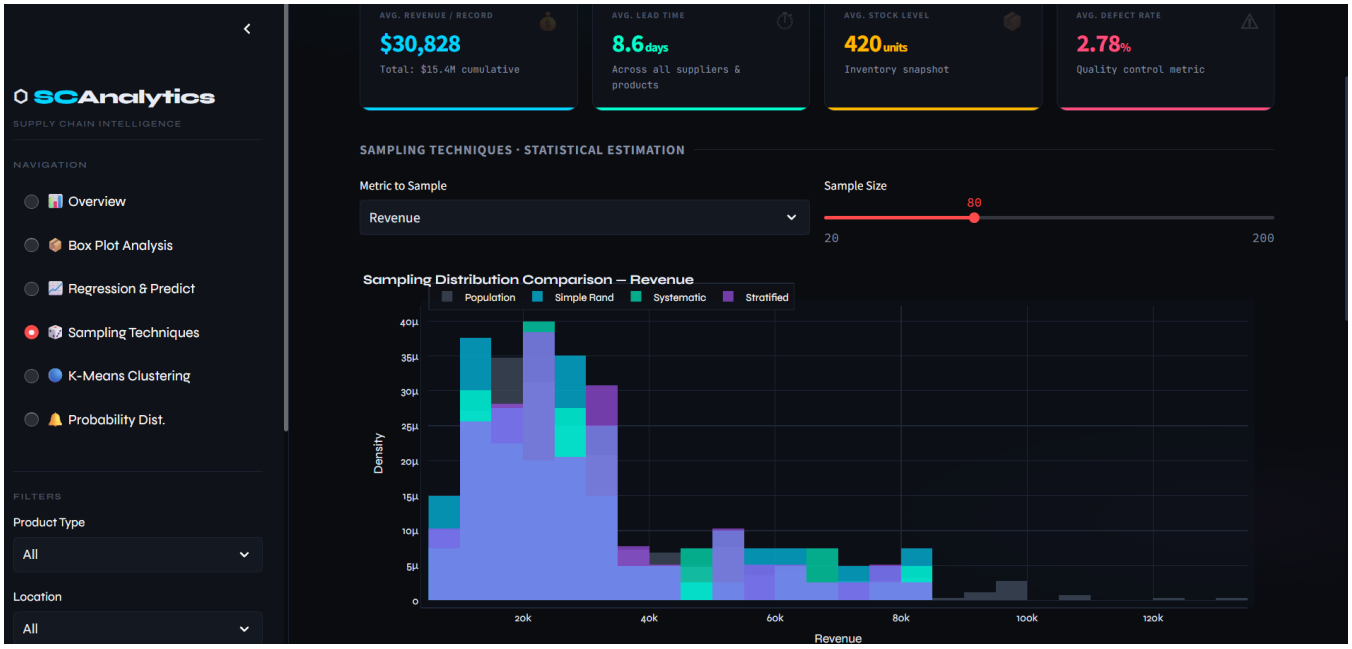
The Regression and Prediction section applies simple linear regression between a selected predictor variable and target variable. The chart shows the relationship between the variables along with a fitted regression line.

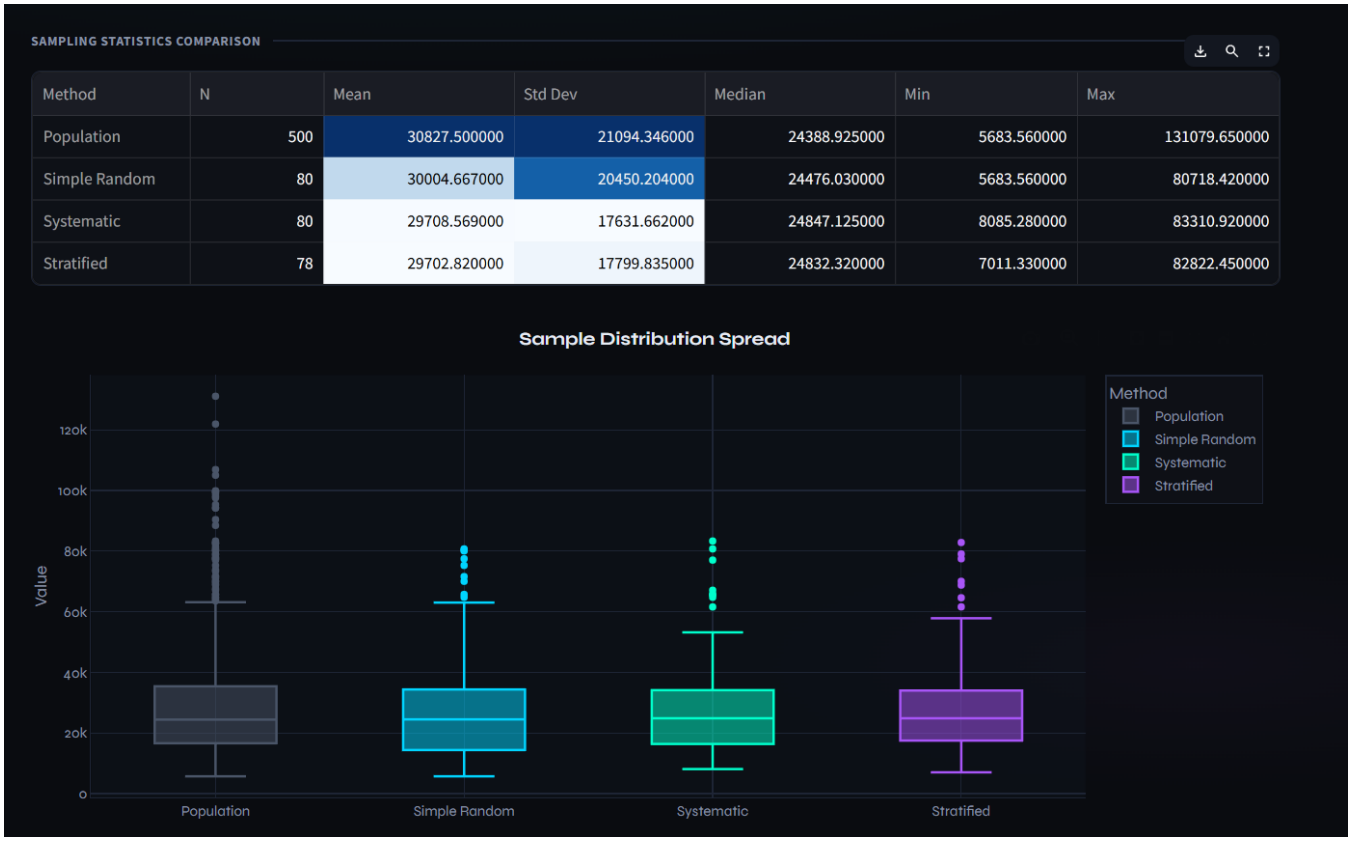
The section also includes a residual plot, regression equation, R-squared score, correlation coefficient, and prediction input. This allows users to explore how strongly two metrics are related and estimate a target value from a selected input.

Key Observations

- The scatter plot and regression line show the direction and strength of the relationship.
- The residual plot helps judge whether the linear model fits the data well.
- The R-squared value explains how much variation is captured by the model.
- The prediction box makes the model interactive and useful for quick estimation.

8. Sampling Techniques





Method	N	Mean	Std Dev	Median	Min	Max
Population	500	30827.500000	21094.346000	24388.925000	5683.560000	131079.650000
Simple Random	80	30004.667000	20450.204000	24476.030000	5683.560000	80718.420000
Systematic	80	29708.569000	17631.662000	24847.125000	8085.280000	83310.920000
Stratified	78	29702.820000	17799.835000	24832.320000	7011.330000	82822.450000

Figure 8: Sampling Techniques

Explanation

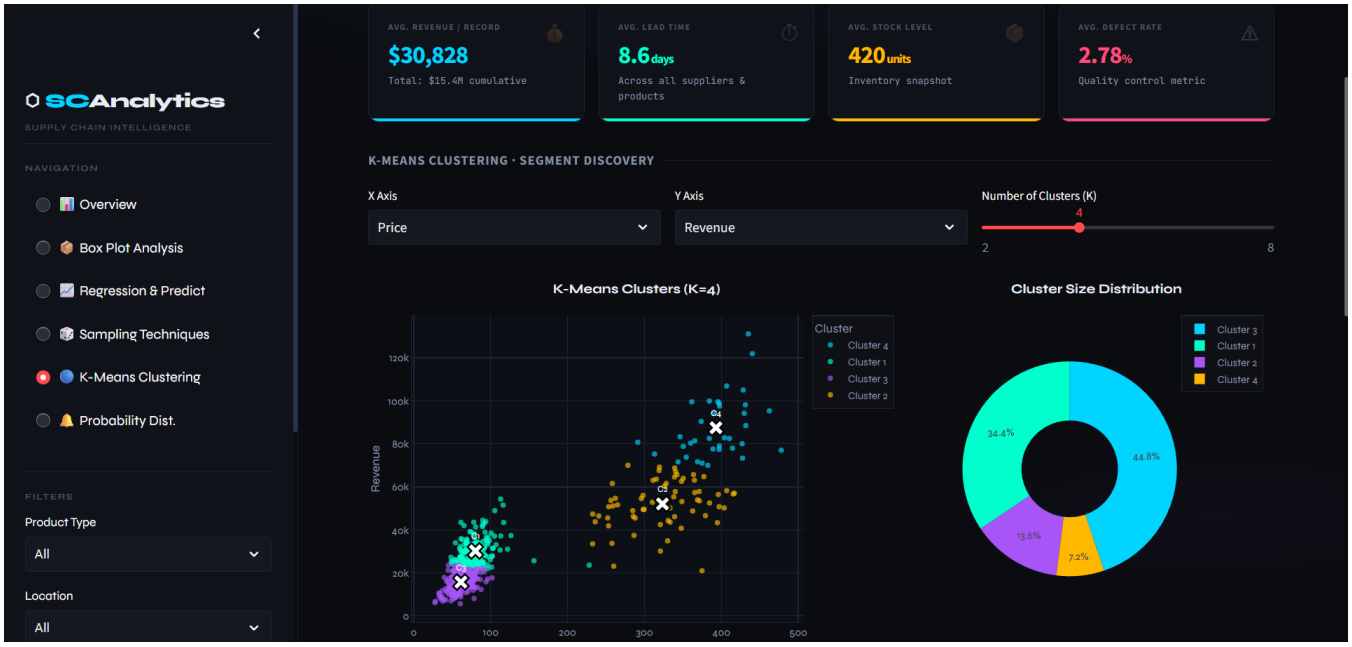
The Sampling Techniques section compares simple random sampling, systematic sampling, and stratified sampling. Users can select a metric and sample size, and the dashboard compares each sample with the full population.

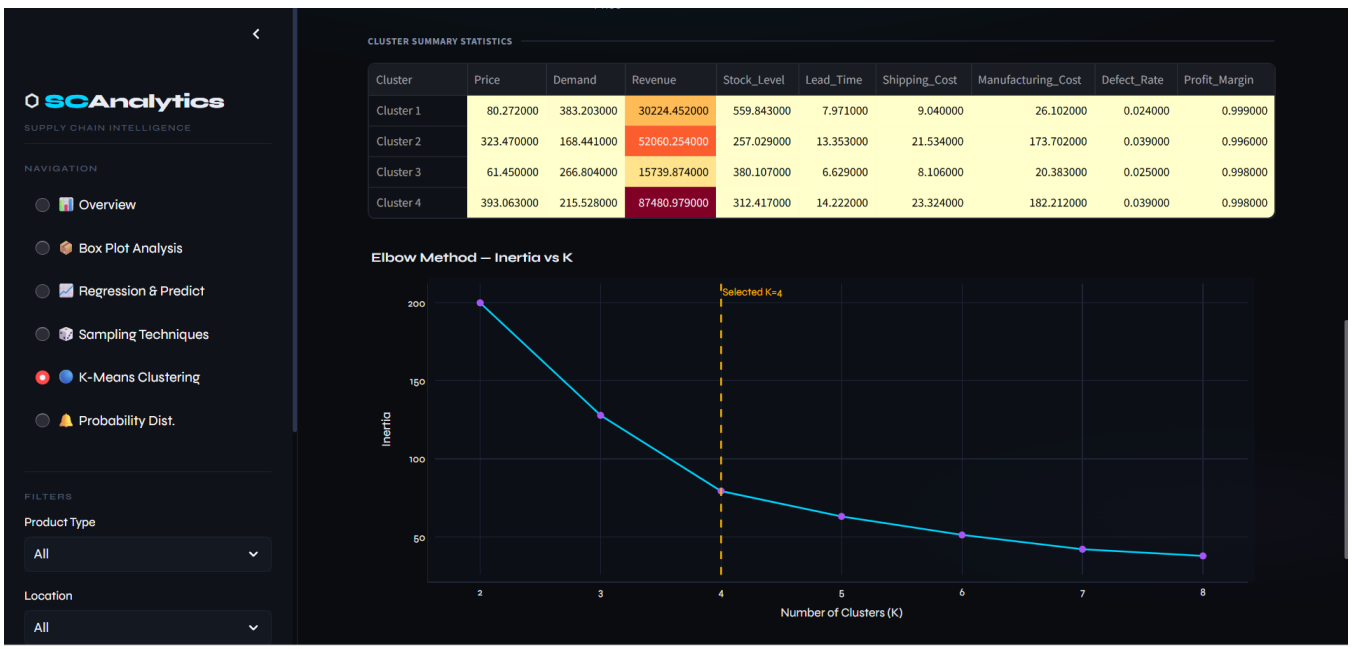
The section includes distribution comparison, sampling statistics, box plot comparison, and bias interpretation. It demonstrates how different sampling methods may produce different levels of accuracy depending on the selected variable.

Key Observations

- The histogram comparison shows how closely each sample follows the population distribution.
- The statistics table compares mean, standard deviation, median, minimum, and maximum values.
- The box plot compares spread and variation across sampling methods.
- The bias result identifies which sampling method gives the closest mean estimate.

9. K-Means Clustering





Cluster	Price	Demand	Revenue	Stock_Level	Lead_Time	Shipping_Cost	Manufacturing_Cost	Defect_Rate	Profit_Margin
Cluster 1	80.272000	383.203000	30224.452000	559.843000	7.971000	9.040000	26.102000	0.024000	0.999000
Cluster 2	323.470000	168.441000	52060.254000	257.029000	13.353000	21.534000	173.702000	0.039000	0.996000
Cluster 3	61.450000	266.804000	15739.874000	380.107000	6.629000	8.106000	20.383000	0.025000	0.998000
Cluster 4	393.063000	215.528000	87480.979000	312.417000	14.222000	23.324000	182.212000	0.039000	0.998000

Figure 9: K-Means Clustering

Explanation

The K-Means Clustering section groups records into clusters based on two selected numerical variables. Users can select the X-axis variable, Y-axis variable, and the number of clusters.

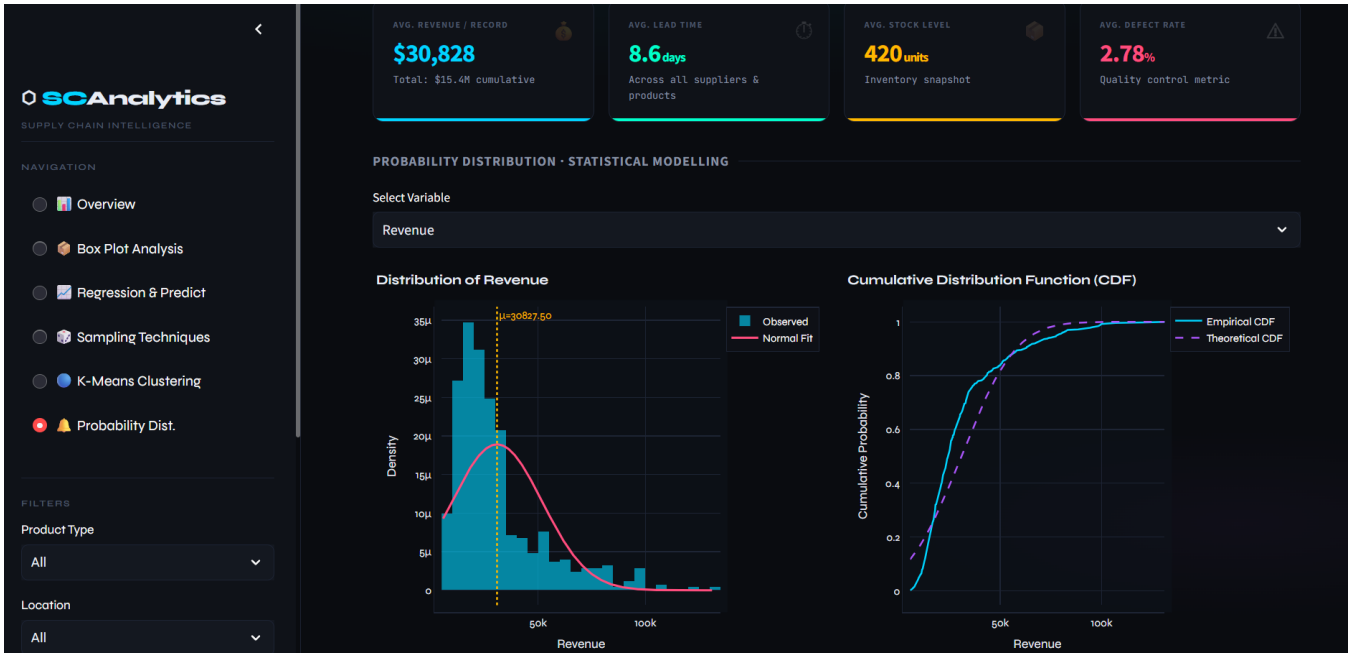
The dashboard displays the clustered scatter plot, centroid markers, cluster size distribution, cluster summary statistics, and elbow method chart. This helps identify groups of records with similar operational or financial behaviour.

Key Observations

- The cluster scatter plot shows how records are grouped based on selected metrics.

- Centroid markers represent the centre point of each cluster.
- The cluster distribution chart shows whether clusters are balanced or uneven.
- The elbow chart helps judge whether the selected number of clusters is suitable.

10. Probability Distribution



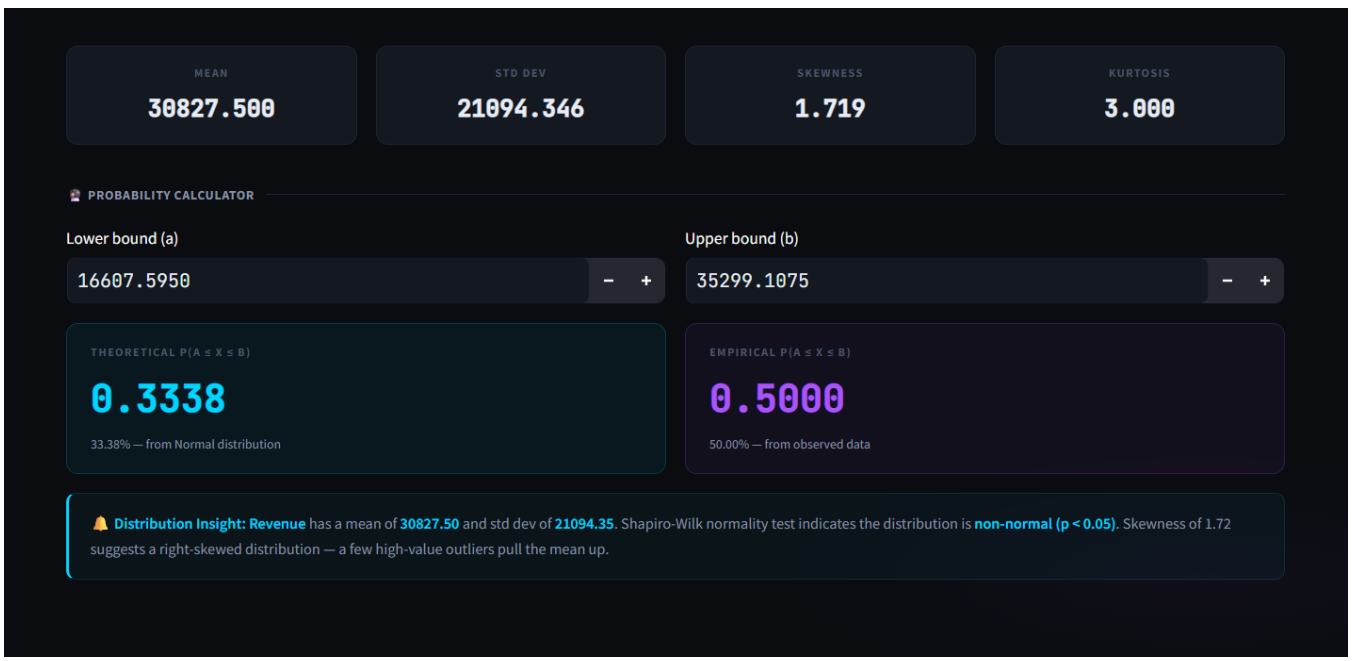


Figure 10: Probability Distribution

Explanation

The Probability Distribution section studies the distribution of a selected numerical metric. It displays a histogram with a normal curve and compares the empirical cumulative distribution with the theoretical cumulative distribution.

The module also reports mean, standard deviation, skewness, kurtosis, and a normality test interpretation. A probability calculator estimates the chance of values falling within a selected interval.

Key Observations

- The histogram shows the observed shape of the selected metric.
- The normal curve helps compare observed behaviour with a theoretical model.
- The CDF chart shows cumulative probability across the metric range.
- The probability calculator converts statistical analysis into an easy interval-based result.

11. Conclusion

The Supply Chain Analytics Dashboard successfully combines descriptive analytics, statistical methods, and machine learning techniques into one interactive tool. The dashboard gives users a clear way to study supply chain performance, compare important business metrics, investigate outliers, build simple predictive models, compare sampling methods, identify clusters, and interpret probability behaviour.

Overall, the project demonstrates how Python, Streamlit, Plotly, Scikit-learn, and SciPy can be used together to create a practical analytics application for supply chain decision support.